

Gamers Nexus, in collaboration with Level1Techs and independent cybersecurity researchers, released a 135-minute investigative report detailing data-tracking practices and security vulnerabilities in LG smart TVs (including retail OLED models like the G5 running webOS).

The report focuses on how LG collects user data and highlights critical hardware and software vulnerabilities.

Key Investigation Findings

- **Continuous Network Scanning & Cross-Device Profiling:** Using packet captures (Wireshark) and firmware analysis, the team discovered that LG smart TVs continuously sweep local area networks (LAN). The TVs identify secondary devices—such as smartphones, PCs, smartwatches, and smart home hardware—mapping their IP addresses and gathering nearby Wi-Fi network SSIDs and signal strengths.
- **LG Ad Solutions Synchronization:** This network mapping feeds into *LG Ad Solutions*, the company's targeted advertising division. While LG reports ~216 million global smart TV sales, LG Ad Solutions claims access to **363 million secondary addressable devices** in the US alone by indexing non-TV household hardware on the same network.
- **Aggressive Automatic Content Recognition (ACR):** The TVs run ACR (via system features like "Live Plus") to sample video/audio and construct digital fingerprints. This logs viewing behavior across all inputs—including broadcast TV, streaming apps, and connected HDMI devices.
- **Audio Capture in Standby Mode:** Researchers demonstrated that the TV's integrated microphone can log ambient audio while the display appears turned off in standby mode. If network access is cut off (e.g., unplugged Ethernet/Wi-Fi), the TV stores the recorded audio locally on the filesystem and silently uploads the cached files once the internet connection is restored.
- **Remote Code Execution (RCE) Flaws:** The investigation uncovered unpatched RCE vulnerabilities in webOS. An attacker who exploits the TV could gain root access, using it as a permanent persistence node to monitor room audio or pivot into other devices across the victim's internal home network.

Discussion & Industry Impact

1. The "Dumb Display" Resurgence & IoT Isolation

The primary technical countermeasure recommended by Gamers Nexus and security analysts is to treating smart TVs strictly as "dumb displays". Users are advised to **disconnect the TV from local Wi-Fi/Ethernet entirely** and rely on external streaming devices (e.g., Apple TV, dedicated media boxes). For users who require native TV app features, placing the TV on an isolated VLAN or Guest IoT network is recommended to prevent local network scanning.

2. Shift from "Ad-Supported" to "Active Surveillance" Business Models

The revelation that LG tracks non-TV secondary devices explains how modern TV manufacturers monetize hardware at low margins. The business model has shifted from selling displays to generating recurring revenue through behavioral ad profiling across a household's entire local network.

3. Illusion of Privacy Toggles & EULAs

The report highlighted that disabling opt-in telemetry or ACR settings in webOS often fails to completely stop background network scanning or local telemetry collection. Complex End User License Agreements (EULAs) bury broad consent clauses for data gathering, leaving consumers with little functional choice if they want to use basic smart functionality.

4. Responsible Disclosure vs. Consumer Risk

Because the uncovered RCE flaws and standby audio logging capabilities present severe network intrusion risks, the team initiated responsible disclosure processes with LG. The fallout has intensified pressure on Smart TV vendors and regulatory bodies (like the FTC) regarding data privacy, local network sniffing, and hardware-level microphone toggles.